Slip 19

Q1) A What is Digital Evidence? Explain the role and purpose of Write Blockers, FTK Imager, and EnCase Imager in forensic acquisition..

1. Digital Evidence

Digital evidence is any information stored or transmitted in digital form that can be used during an investigation or in a court of law.

Examples:

- Emails and chat messages
- Computer files and documents
- Browser history
- Call logs and mobile data
- Images, videos, and network logs

2. Write Blocker

A write blocker is a hardware or software tool that prevents data from being written to a digital storage device during forensic examination.

Role and purpose:

1. Prevents accidental modification of the original evidence.
2. Allows investigators to read data without changing it.
3. Maintains the integrity and reliability of digital evidence.
4. Helps preserve the original storage device for investigation.

Example: Connecting a suspect's hard disk through a hardware write blocker before creating a forensic image.

3. FTK Imager

FTK Imager is a digital forensic tool used to create forensic images and examine digital evidence.

Role and purpose:

1. Creates forensic copies of hard disks, USB drives, and other supported media.

2. Allows investigators to preview files and folders.
3. Supports examination of deleted files and file-system information where available.
4. Can calculate hash values, such as MD5 and SHA-1, to help verify data integrity.
5. Saves forensic images in supported formats, such as E01 and raw formats.

Example: Creating a forensic image of a USB drive and calculating its hash value to help verify the image.

4. EnCase Imager

EnCase Imager is a forensic acquisition tool associated with EnCase forensic products. It is used to acquire and preserve digital evidence for further examination.

Role and purpose:

1. Creates forensic images of hard disks and other supported storage media.
2. Supports acquisition of digital evidence while preserving the original source.
3. Helps verify acquired data using hash values.
4. Records acquisition-related information for documentation.
5. Produces forensic image files that can be examined using compatible forensic tools.

Example: Creating a forensic image of a suspect's hard disk for later analysis in a digital investigation.

5. Difference Between These Tools

Basis	Write Blocker	FTK Imager	EnCase Imager
Main purpose	Prevents writing to evidence media	Acquires and previews evidence	Acquires and preserves evidence

Type	Hardware or software protection tool	Forensic imaging tool	Forensic imaging tool
Data imaging	Not its primary purpose	Yes	Yes
Evidence protection	Prevents writes through the blocker	Supports read-only acquisition workflows	Supports forensic acquisition workflows
Hash verification	Not its primary purpose	Supported	Supported
Example	Protecting an original USB drive	Creating a USB forensic image	Creating a hard-disk forensic image

6. Importance in Digital Forensics

- Integrity: Helps ensure evidence remains unchanged.
- Accuracy: Creates forensic copies for examination.
- Verification: Hash values help detect changes to acquired data.
- Documentation: Acquisition records support the chain of custody.
- Legal reliability: Proper procedures help establish the authenticity and reliability of evidence.

Conclusion

Digital evidence is important for investigating cybercrimes. A write blocker protects original storage media from writes, while FTK Imager and EnCase Imager help create forensic images for examination. Together with proper

documentation, hashing, and chain-of-custody procedures, these tools support reliable digital forensic investigations.

Q2) B Write a Python program that accepts a sentence from the user and encrypts it using Caesar Cipher

```
text = input("Enter a sentence: ")

key = int(input("Enter key: "))

enc = ""

for ch in text:

    if ch.isalpha():

        enc = enc + chr((ord(ch.upper()) - 65 + key) % 26 + 65)

    else:

        enc = enc + ch

print("Encrypted text:", enc)
```

Q2) B Write a program to implement RSA algorithm.(Generate public and private keys, encrypt a message, and decrypt it.)

```
p = 3
q = 11
n = p * q
phi = (p - 1) * (q - 1)

e = 7
d = 3

print("Public key:", e, n)
print("Private key:", d, n)

M = int(input("Enter message number (0 to 32): "))

# Encryption
C = (M ** e) % n
```

```
print("Encrypted message:", C)

# Decryption
M = (C ** d) % n
print("Decrypted message:", M)
```